

Pedogenic Transformation of Fractured Granitic Bedrock, Southern California

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ABSTRACT

Joint fractures in granitic bedrock are known to influence patterns of weathering and landform evolution, but their influence on pedogenic processes has not been thoroughly studied. This research was designed to elucidate dominant pedogenic processes and the role of joint fractures in the transformation of a soil-weathered bedrock profile in the San Jacinto Mountains of southern California. The granitic bedrock is sufficiently weathered to meet paralithic materials criteria, but is not so intensively weathered to be considered saprolite. A lateral sequence of five morphologic zones exists within each weathered bedrock horizon: the matrix, matrix rind, fracture rind, fracture coating, and fracture fill. Distributions of pedogenic clay and citrate-bicarbonate-dithionite extractable Fe (Fe_d), and micromorphologic observations of mineral grain alteration indicate that the effects of chemical weathering increase upward in the profile and laterally toward joint fractures. Illuvial clay was detected throughout the profile, but is most abundant in the shallowest weathered bedrock horizons and in morphologic zones near fracture margins. Shrink-swell activity of clay deposited in planar voids, microcracks, and root channels is responsible for mineral grain fragmentation and localized disruption of rock-controlled fabric. Stresses generated by swelling clay are also responsible for fabric reorganization and development of incipient subsoil structure in the shallowest weathered bedrock horizon. Illuviation and shrink-swell processes play critical roles in the pedogenic transformation of this granitic bedrock.

GRANITIC BEDROCK underlies $\approx 20\%$ of California's land area, concentrated primarily in the Sierra Nevada and the southern California Peninsular Mountain Ranges (Norris and Webb, 1990). Where unaffected by Pleistocene glaciation, granitic bedrock is often weathered to depths of several meters, while overlying soils are comparatively thin (Wahrhaftig, 1965; Graham et al., 1997). Because of its widespread occurrence, not only in California, but throughout much of the world (Gerrard, 1994), weathered granitic bedrock is an important soil parent material. As such, its properties and the processes by which it transforms to soil are of interest from a pedological standpoint. Because weathered bedrock often dominates granitic regoliths, an understanding of its functions in natural systems is of interest relative to ecology, hydrology, environmental quality, and engineering as well.

The roles of weathered granitic bedrock in soil genesis, water movement, and natural ecosystems have been studied worldwide (Isherwood and Street, 1976; Koppi and Williams, 1980; Megahan and Clayton, 1986; Evans and Bothner, 1993; Graham et al., 1994). Several studies have focused on granitic bedrock in the western USA.

Nettleton et al. (1968, 1970) studied selected chemical, physical, mineralogical, and morphological properties of weathered granitic bedrock and overlying soils on a toposequence in the southern California Peninsular Ranges and used their data to determine processes of bedrock weathering and soil formation. In the Idaho batholith, Clayton and Arnold (1972), Clayton et al. (1979), and Clayton (1993) characterized and discussed various properties and processes in granitic bedrocks during different stages of weathering and developed a practical classification scheme. Weathered granitic bedrock in the San Jacinto Mountains in southern California has been shown to possess abundant lithogenic and pedogenic porosity, and to conduct water relatively rapidly in comparison with overlying soils (Johnson-Maynard et al., 1994; Graham et al., 1997). The ecological significance of weathered granitic bedrock in southern California and the southern Sierra Nevada has been demonstrated by studies showing that it serves as a rooting medium and as a source of plant available water for forest and chaparral vegetation (Arkley, 1981; Jones and Graham, 1993; Anderson et al., 1995; Sternberg et al., 1996; Hubbert, 1999).

The granitic bedrocks described in the previously cited studies, as with most granitic bedrocks, possess joint fractures produced by the release of geostatic and tectonic forces over time (Ollier, 1969; Twidale, 1971). Joint fractures facilitate the ingress of weathering agents such as water and plant roots, influence patterns of weathering and clay illuviation, and affect landform evolution (Ollier, 1965; Twidale, 1971; Clayton et al., 1972; Graham et al., 1994; Ollier and Pain, 1996); however, the influence of fractures on specific pedogenic processes and their role in the transformation of granitic bedrock to soil have not been thoroughly studied.

The objectives of this study were (i) to characterize various chemical, physical, morphologic, and micromorphologic properties of a weathered granitic bedrock in southern California, with an emphasis on properties near joint fractures, and (ii) to use the data collected during this process to interpret dominant pedogenic processes and the overall role of joint fractures in the transformation of the fractured granitic bedrock to soil.

MATERIALS AND METHODS

Study Area and Site Characterization

The study area is at an elevation of 1150 m in the northwestern foothills of the San Jacinto Mountains in southern California, ≈ 8 km south of the town of Banning. The San Jacinto mountains are part of the larger southern California Peninsu-

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Abbreviations: COLE, coefficient of linear extensibility; Fe_d , citrate-bicarbonate-dithionite extractable Fe; OC, organic C.

lar Range complex (Norris and Webb, 1990), and are composed mostly of plutonic igneous rocks with lithologies ranging from mafic tonalite to K-feldspar-poor granodiorite. Compositional variation is minimal and ranges from 50 to 55% plagioclase (An_{30-40}), 20 to 30% quartz, 1 to 8% K-feldspar, 10 to 15% biotite, and 0 to 5% hornblende, with mineral grain size usually <10 mm (Hill, 1988). The area is characterized by a Mediterranean climate, with precipitation occurring mostly as rain between the months of November and April. The closest weather stations are located in Idyllwild (20 km southeast, elevation 1640 m), which receives ≈ 645 mm of precipitation annually, and in Beaumont (12 km northwest, elevation 792), which receives ≈ 432 mm of precipitation annually (National Oceanic and Atmospheric Administration, 1992). Soils in the area consist of moderately deep to shallow Entisols, Alfisols, and Mollisols on gently rolling to steep hillslopes (Knecht, 1971; Graham et al., 1997). Vegetation consists of chaparral species dominated by chamise (*Adenostoma fasciculatum* Hook. & Arn.), Eastwood's manzanita (*Arctostaphylos glandulosa* Eastw.), and cupleaf ceanothus [*Ceanothus greggii* A. Gray var. *perplexans* (Trel.) Jep.].

The site investigated is on a nearly level ridge top. Soils are loamy, mixed, active, mesic, shallow Typic Haploxeralfs underlain by >1 m of weathered granitic bedrock. The regolith materials are moderately acid, with pH values ranging from 4.95 in the uppermost soil horizon to 5.76 in the deepest weathered bedrock horizon studied (1:1 soil/water) (Frazier, 1997). As reported by Frazier (1997), the primary mineral composition of the bedrock at a depth of 1.25 m is 40% plagioclase feldspar, 21% quartz, 20% mica (mostly biotite), 13% K-feldspar, and 5% hornblende. The fine sand (100–250 μm) and medium silt (5–20 μm) fractions of the weathered bedrock horizons above a depth of 1.25 m contain quartz, plagioclase, hornblende, and biotite, and the secondary minerals kaolin, vermiculite, and regularly and randomly interstratified mica-vermiculite. The clay fractions (<2 μm) of the weathered

bedrock and overlying soil horizons contain no primary minerals, and secondary minerals are identical to those contained in the coarser size fractions, with the addition of trace quantities of goethite.

Less mafic tonalites (<1% hornblende, $\approx 10\%$ biotite) occur in the general study area, but produce soil and weathered bedrock morphologies that differ from those described in this study. In particular, much less illuvial clay is generated (Graham et al., 1997).

Field Methods

A trench 1.25 m deep, 2.25 m long, and 0.75 m wide was excavated using an electric chipping hammer and hand implements. The weathered bedrock at the study site and throughout the general area was extensively fractured. The widest and longest fractures exposed on the trench walls were vertically oriented joints — hereafter referred to as joint fractures — which were usually continuous from the soil–weathered bedrock interface to the lowermost exposed bedrock (1.25 m).

Alteration of the original bedrock material has produced a vertical sequence of weathered bedrock horizons and a series of morphologic zones emanating laterally from the fractures. Based on these observations, each weathered bedrock horizon was divided into four zones for description and sampling: the matrix, fracture rind, fracture coating, and fracture fill (Fig. 1). For the purposes of sampling, the matrix was considered to be the material located >2 cm from the fracture sidewalls. The fracture rind consisted of more intensely altered bedrock material that extended from the fracture sidewall to ≤ 2 cm into the matrix. The fracture coating was a thin (<0.1 cm) layer of material covering the surface of the fracture rind. The fill material was contained within the fracture itself, and in most cases, did not completely fill the fractures. An additional zone, the matrix-rind zone, was not identified in the field, but was discovered and described during micromorphological

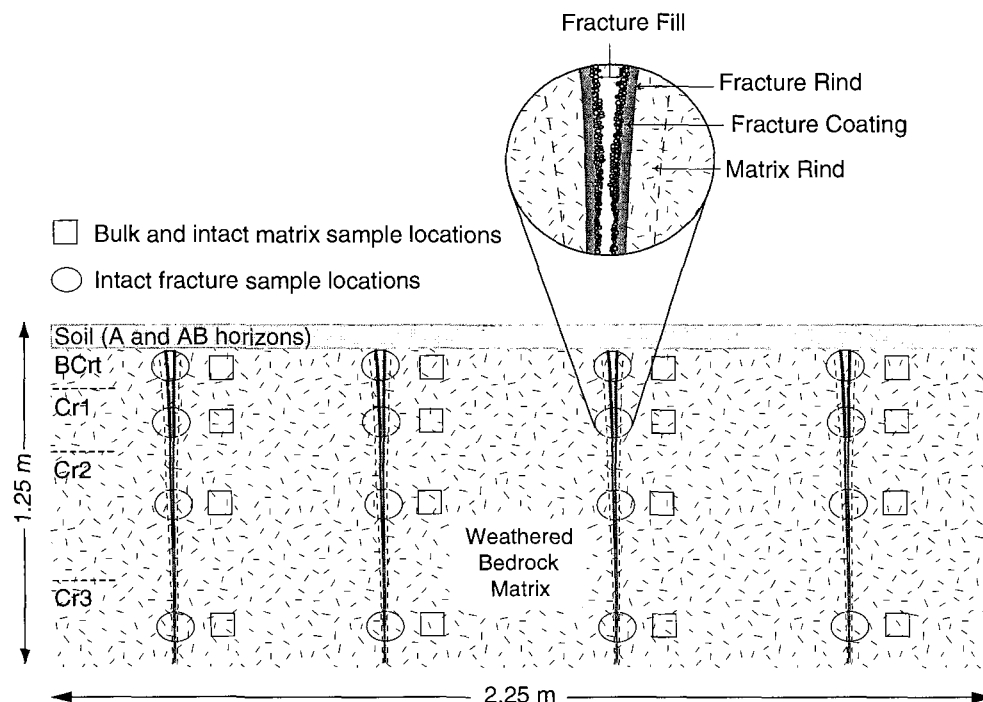


Fig. 1. Schematic representation of weathered bedrock morphology at the study site showing the location and frequency of intact and bulk samples, and the relative positioning of the five morphologic zones (fill, coating, rind, matrix rind, and matrix) surrounding a typical joint fracture. Fracture dimensions are approximately to scale.

examination (Fig. 1). This zone had a variable, undetermined width and consisted of matrix material immediately adjacent to the fracture rind. It was less altered than material in the fracture rind, but more altered than matrix material located farther from joint fractures.

A detailed morphologic description of the soil and underlying weathered bedrock was made on the trench exposure, which included four joint fractures (Fig. 1). The fractures were on average 63 cm apart and were continuous from the soil-weathered bedrock interface (7-cm depth) to the bottom of the trench (125 cm). Intact samples of weathered bedrock, ≈ 10 to 15 cm in diam., were collected from each horizon in the region of the bedrock fractures using the sampling scheme shown in Fig. 1. Each sample contained the complete lateral morphologic zonation of matrix through fracture fill. Additional bulk and intact samples were collected from the weathered bedrock matrix at positions 10 to 20 cm from the fracture sidewalls (Fig. 1), and used for all analyses of the matrix materials. Bulk and, when possible, intact samples were also collected from the overlying soil horizons.

Laboratory Methods

Materials from the four different fracture morphologic zones were separated from the intact sample blocks using a metal spatula. Soil and weathered bedrock samples were air dried, crushed, and sieved to remove coarse fragments (>2 mm). Unless otherwise noted, all subsequent analyses were performed on the fine earth fraction (<2 mm). Particle-size distribution was determined by the pipette method after organic matter removal using H_2O_2 (Gee and Bauder, 1986). Total C was measured with a Carlo Erba NA1500 C/N/S analyzer (Fisons Instruments, Dearborn, MI) by measuring CO_2 evolved during dry combustion at a temperature of $1029^\circ C$ (Nelson and Sommers, 1982). Total C was assumed equal to organic C (OC) because carbonates were not detected in the soil or weathered bedrock. Bulk density was measured on intact samples using the paraffin-coated clod method (Blake and Hartge, 1986b). Particle density was measured using an Accupyc 1330 gas pycnometer (Micromeritics Instr. Corp., Norcross, GA) (Blake and Hartge, 1986a). Total porosity was calculated from bulk density and particle density. Samples were extracted with citrate-bicarbonate-dithionite to determine quantities of pedogenic Fe (Jackson et al., 1986). Soil and weathered bedrock color were measured with a Minolta CR-200 Chroma Meter (Minolta Corp., Ramsey, NJ). Coefficient of linear extensibility (COLE) values were determined for the weathered bedrock matrix from the bulk density of Saran-coated clods desorbed to -10 KPa and oven dryness (Method 4D; Soil Survey Staff, 1984). Data from the particle size, bulk density, OC, and Fe_d procedures were analyzed statistically using two-factor ANOVA, with the factors being horizon (depth) and morphologic zone. Multiple comparisons were made using Tukey's least significant difference procedure (Data Desk, 1995).

Thin sections (25 by 45 mm) were prepared from intact samples of the matrix and fracture-rind zones from each of the four weathered bedrock horizons. Thin sections were viewed with a petrographic microscope and described according to the terminology of Bullock et al. (1985). When referring to illuvial clay features, the terminology of Brewer (1976) was also used. Point counting procedures were used to estimate the volume percentages of channels and planar voids with diameters ≥ 0.1 mm (i.e., macropores; Vepraskas et al., 1991), and void coatings and infillings. Estimates were based on a total of 300 counts per morphologic zone (spread equally across three slides).

Table 1. Morphologic description of the soil and underlying weathered bedrock at study site.

Table 1. Morphologic description of the soil and underlying weathered bedrock at study area.														
Horizon	Depth (cm)	Matrix morphology					Fracture morphology							
		Boundary†	Dry color	Texture	Structure‡	Dry consistence§	Roots¶	Clay films#	Dry rind color	Rind texture	Rind thickness		Fracture aperture	Roots‡‡
											Range	Mean††		
A	0-2	as	9.3YR5.0/3.5	grsl	3cpl	sh	2vlf lm,c		--§§	-	-	-	-	-
AB BCrt	2-7 7-17	aw cw	9.2YR4.9/3.3 7.8YR5.1/3.8	vgrsl grsl	lmsbk → lfsbk	so vh	3vlf 2vlf,m lc	1n,mk	7.3YR4.6/3.5	grsl	-	0.6-2.2	1.3 ^a	1.0-1.5
	17-45	cw	9.1YR5.6/3.4	grsl	rock structure	vh	lf,m		8.3YR4.7/3.4	grsl		0.5-1.7	0.9 ^b	0.8-1.2
Cr1	17-45													
Cr2	45-88	gw	9.9YR5.6/3.4	vgrsl	rock structure	h-vh	nd§§		9.6YR5.1/3.3	grsl		0.1-1.1	0.5 ^c	0.4-0.6
Cr3	88-125+		0.7Y5.9/2.9	vgrsl	rock structure	h-vh	nd		9.5YR5.3/3.4	grsl		≤0.5	0.3 ^d	≤0.4
														2lvf lm,c

[†] a = abrupt (0.5- <2 cm), c = clear (2-5 cm), g = gradual (5-15 cm); s = smooth, w = wavy.

[‡] 3 = strong, 1 = weak; c = coarse, m = medium, f = fine; pl = platy, sbk = subangular blocky; → = parting to.

[§] sh = slightly hard, so = soft, vh = very hard, h = hard.

^{||} 3 = many, 2 = common, 1 = few; vf = very fine, f = fine, m = medium, c = coarse.

[#] 1 = few; n = thin; mk = moderately thick.

^{††} Values in column with different superscripts differ significantly ($P \leq 0.05$).

^{‡‡} Includes roots contained within the rind and roots contained within the fracture void space.

§§ = not applicable; nd = none detected.

RESULTS

Morphology

Selected morphologic properties of the soil and underlying weathered bedrock appear in Table 1. The weathered bedrock matrix composed most of the regolith and exhibited distinct horizonation with depth. Roots were distributed throughout the BCrt matrix, but were much less abundant than in overlying soil horizons. Below the BCrt horizon, the matrix consisted of a series of Cr horizons that showed no structural development and differed from one another largely on the basis of color, with hue becoming progressively yellower with increasing depth (Table 1). Root growth in the matrix of the Cr horizons was minimal, occurring only in the upper third of the Cr1 horizon. Below the Cr1 horizon, roots in the matrix occurred primarily in small lateral fractures leading away from larger joint fractures or in regions of the matrix ≤ 5 cm from fracture sidewalls.

The aperture of vertical joint fractures decreased with depth, from 1.5 cm in the BCrt horizon to ≤ 0.4 cm in the Cr3 horizon (Table 1). These measurements were made on the trench face after it had been exposed to temperatures of $\approx 20^\circ\text{C}$ for ≈ 10 h. The width of the fracture rind also decreased with increasing depth, from 0.6 to 2.2 cm in the BCrt horizon to ≤ 0.5 cm in the Cr3 horizon. At most depths, the rind was distinguished from the adjacent matrix largely by a contrast in color (Table 1). The color contrast, and thus the boundary between the two zones, was abrupt at all depths, but was much less pronounced in the BCrt horizon than in the Cr horizons. At all depths, roots contained within the fracture void space were concentrated along the fracture sidewalls. Few of these roots extended laterally into the adjacent fracture-rind and matrix zones.

Fracture-coating materials in the BCrt, Cr1, and Cr2 horizons closely resembled pressure faces commonly found on the ped surfaces of clayey subsoils. Many of the sand-sized mica particles in these coating materials exhibited parallel-referred orientation patterns relative to the surfaces of the fracture rind, giving the rind surface a shiny appearance. Coating materials in the Cr3 horizon, and sometimes those in the Cr2 horizon, had

a dull, powdery appearance, and more closely resembled illuvial silt films than pressure faces.

Fracture-fill materials were found in the void space of all four fractures studied. They were usually structureless with a loose or soft consistency, but occasionally had a compact platy structure oriented parallel to the vertical axes of the joint fractures. In most cases, the fill materials adhered somewhat to the fracture sidewalls.

Qualitative Micromorphology

Weathered Bedrock Matrix

The fabric of the matrix in all four weathered bedrock horizons was rock controlled, in that the distribution and orientation of mineral grains and void spaces was largely controlled by structural and textural properties of the bedrock (Stolt et al., 1991). Chemical mineral weathering increased upward in the profile. Dissolution pitting of feldspar minerals occurred as linear alterations along twin planes and as complex alteration patterns spread randomly across grain surfaces (Fig. 2a). The dissolution pits contained secondary weathering products, including Fe oxides, which imparted a dirty appearance on grain surfaces (Fig. 2a). Parallel linear alteration of biotite grains was apparent at all depths studied (Fig. 2a). More advanced stages of biotite weathering, as indicated by yellower grain colors and decreased birefringence and pleochroism, become more apparent upward in the profile. Hornblende alteration was weak and was apparent only from accretions of Fe oxides along cleavage planes and in pores contiguous with grain boundaries. Quartz grains appeared chemically unaltered, but exhibited abundant intragranular microcracks throughout the profile (Fig. 2a).

Macroporosity and mesoporosity consisted of root channels and inter-, intra-, and transmineral planar voids. Transmineral and intermineral planar voids occurred across all grain types, while intramineral planar voids occurred mainly as microcracks in quartz grains and in between expanded mica sheets. The basic and related orientation patterns of planar voids were generally random; however, many planar voids exhibited ra-

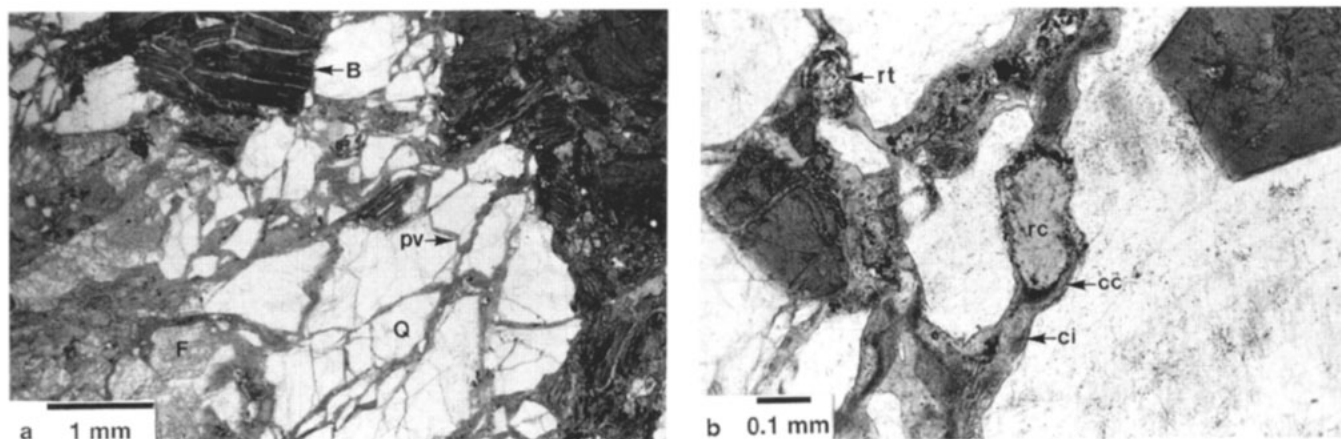


Fig 2. Photomicrograph (plane polarized light) of (a) the Cr1 matrix, showing rock-controlled fabric, parallel linear alteration of biotite (B), dissolution pitting of feldspars (F), and clustered distribution patterns of angular quartz grains separated by microcracks and planar voids (pv) infilled with illuvial clay; (b) the Cr3 matrix, showing root channels (rc) contained within planar voids, illuvial clay coatings (cc) and infillings (ci), and root tissue (rt).

dial-referred distribution patterns relative to expanded mica grains (Isherwood and Street, 1976).

Coatings and infillings of mostly clay-sized weathering products, including Fe oxides, were present on the surfaces of channels and planar voids at all depths studied (void ferriargillans of Brewer, 1976) (Fig. 2a and

2b). They exhibited banded extinction patterns under cross polarizers, which indicated weak-to-strong parallel-referred orientation patterns relative to void surfaces and suggested an illuvial origin. Void coatings and infillings in the Cr horizons usually had a hue of 10YR, while those in the BCrt horizon had a hue of 7.5YR or redder.

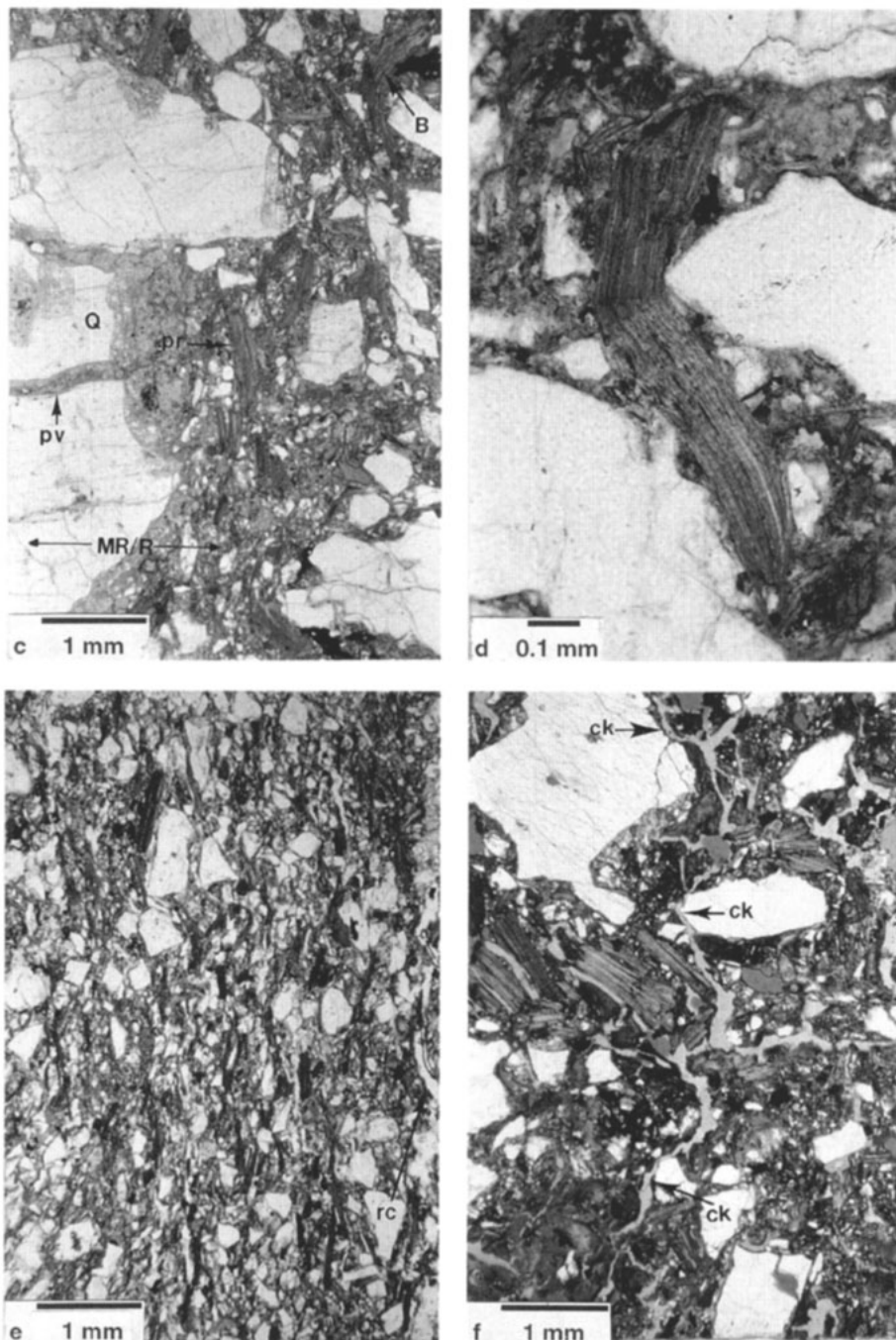


Fig 2 continued. Photomicrograph (plane polarized light) of (c) the boundary between the matrix rind and fracture rind zones in the Cr1 horizon (MR/R), showing the gefuric *c/f* related distribution pattern of the fracture rind, the parallel referred orientation pattern (pr) of platy biotite grains in the fracture rind relative to the vertical axis of the joint fracture, a contorted biotite grain (B) in the fracture rind, and sand-sized quartz grains (Q) at the boundary between the matrix-rind and fracture-rind zones fragmenting along planes of weakness corresponding to infilled planar voids (pv); (d) a bow-like, contorted biotite grain in the Cr1 rind; (e) the most common type of fabric in the BCrt rind, showing the gefuric *c/f*-related distribution pattern, the strong parallel referred orientation pattern of platy biotite and elongate quartz and feldspar grains relative to vertical axis of the joint fracture, and a root channel (rc) running along the boundary between the matrix-rind and fracture-rind zones, (f) the second, less common type of fabric in BCrt fracture rind, showing the gefuric *c/f* related distribution pattern and crack (ck) and incipient subangular blocky microstructures. Note: photomicrographs c, d, e, and f are oriented so that the vertical axis of the joint fractures is parallel with the long axis of the photomicrographs.

Fragmented, sand-sized mineral grains occurred regularly throughout the matrix. The abundance of fragmented mineral grains and the degree of mineral grain fragmentation generally increased upward in the profile. Fragmentation occurred mainly along planes of weakness corresponding to infilled planar voids, an example of which is shown in Fig. 2a. The group of quartz grains in the center of the photomicrograph had a clustered distribution pattern in which each grain in the cluster had boundaries that appeared to have once accommodated adjacent grains. Grain fragmentation patterns such as these typically occur in areas of the matrix where the rock-controlled fabric has been locally disrupted.

Matrix-Rind Zone

When viewed in thin section, the region of the matrix within 1 to 3 cm of the rind had rock-controlled fabric, but had features that contrasted slightly with those in

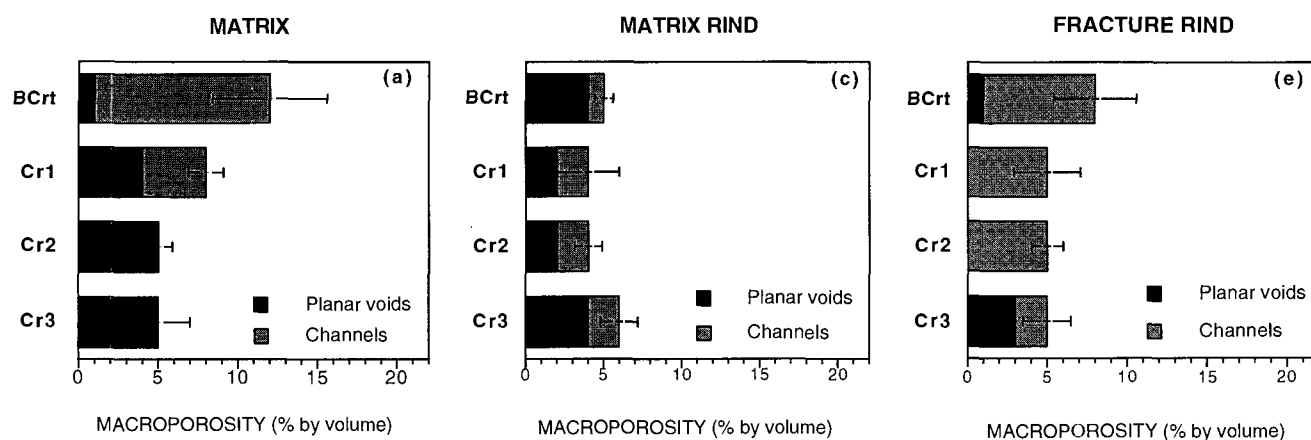
the remainder of the matrix. Because of its position, it was termed the matrix-rind zone. It was evident only under magnification, and was not sampled for analysis of chemical and physical properties.

The degree of chemical mineral weathering in the matrix-rind zone was greater than in the rest of the matrix, especially with regard to the degree of feldspar pitting. The most distinguishing characteristics of the matrix-rind zone, however, were that the abundance of fragmented mineral grains and the degree of mineral grain fragmentation were greater than in the adjacent matrix. As in the matrix, fragmented mineral grains were separated mainly along the axes of infilled planar voids and microcracks (Fig. 2c).

Fracture-Rind Zone

In the Cr3 horizon, the rind fabric was typically rock controlled and therefore very similar to the fabric of

MACROPOROSITY



VOID COATINGS & INFILLINGS

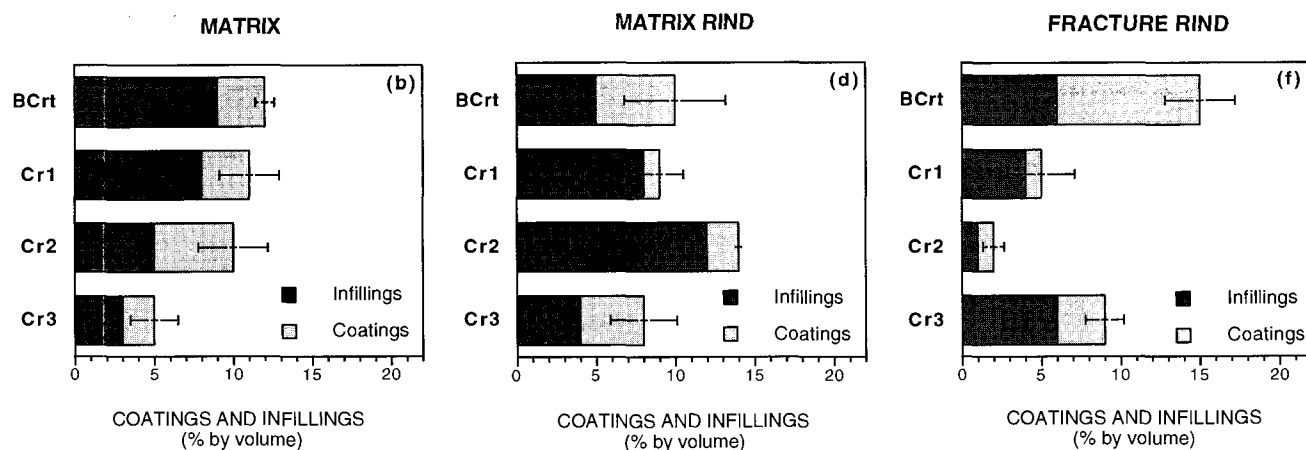


Fig. 3. Relative proportion of macroporosity (root channels + planar voids) and void coatings and infillings in the matrix, matrix rind, and fracture-rind zones of each weathered bedrock horizon. Error bars are for the entire value bar, (i.e., for macroporosity plot, error bar is for the mean value of channels + planar voids), and represent one standard deviation about the mean. Macropore ≥ 0.1 -mm diam.

the adjacent matrix-rind zone; however, fragmented mineral grains were more abundant and the degree of grain fragmentation was more advanced in the rind than in the matrix-rind zone.

In the Cr2 and Cr1 horizons, the rind had gefuric and single-spaced porphyric *c/f*-related distribution patterns and contained more fine materials than in the Cr3 horizon (Fig. 2c). It lacked a definite microstructure, but the parallel-referred orientation pattern of the platy mica grains and elongate quartz and feldspar grains relative to the joint fracture sidewalls gave the rind in the Cr2 and Cr1 horizons a somewhat laminated appearance (Fig. 2c). Platy mica particles with "bow-like" or contorted configurations (Fig. 2c and 2d) were common throughout the rind in the Cr2 and Cr1 horizons, as were granostriated clay and mica pseudomorphs at the boundaries of sand-sized quartz and feldspar grains. The boundary between the rind and matrix-rind zone in both Cr2 and Cr1 horizons was sharp and prominent (at 20 $\times$ magnification; Bullock et al., 1985), and was usually <1 mm wide. The boundary was sharpest where it ran along quartz grains, and was more diffuse where it abutted highly weathered feldspar, mica, and hornblende grains (Fig. 2c).

In the BCrt horizon, the rind exhibited two types of fabric. In the first, which was more common, the *c/f*-related distribution pattern and the referred orientation pattern of platy and elongate mineral grains were similar or identical to those described above for the rind in the Cr2 and Cr1 horizons (Fig. 2e). In the second, the *c/f*-related distribution pattern was also gefuric or single-spaced porphyric, but crack and fissure microstructures were apparent and the parallel-referred orientation patterns of sand-sized mica, quartz, and feldspar grains, which was strongly expressed in the first type of fabric, was weak or nonexistent (Fig. 2f). In some cases, the crack and fissure microstructures appeared together with weakly developed subangular blocky microstructure. In samples with this second type of fabric, the boundary between the rind and matrix rind was usually faint and diffuse (at 20 $\times$ magnification).

Quantitative Micromorphology

Macroporosity

In the matrix, macroporosity increased upward in the profile (Fig. 3a). Planar voids composed most of the macropore space in the matrix of the Cr horizons, whereas root channels composed most of the macropore space in the matrix of the BCrt horizon. In the matrix-rind zone, macroporosity was lower than that in the matrix in the BCrt, Cr1, and Cr2 horizons, and was slightly greater than that in the matrix in the Cr3 horizon (Fig. 3a and 3b). In the fracture rind, macroporosity was about the same as, or slightly higher, than in the matrix rind, and consisted mostly of root channels (Fig. 3b and 3c).

Void Coatings and Infillings

In the matrix, the volume of void coatings and infillings increased upward in the profile (Fig. 3d), a trend

that coincided with a decrease in the volume of planar macropores (Fig. 3a), and an increase in the abundance of fragmented mineral grains and the degree of mineral grain fragmentation. In the matrix-rind zone, the volume of void coatings and infillings was greater than that in the matrix in the Cr3 and Cr2 horizons, and was similar to that in the matrix in the Cr1 and BCrt horizons (Fig. 3d and 3e). In the fracture-rind zone, the volume of void coatings and infillings was significantly lower than that in the adjacent matrix rind and matrix zones in the Cr1 and Cr2 horizons (Fig. 3d, 3e and 3f). Conversely, the volume of void coatings and infillings in the rind was greater than that in the adjacent matrix and matrix-rind zones in the BCrt horizon. In the Cr3 horizon, the volume of void coatings and infillings in the rind was similar to that in the matrix rind, but greater than that in the matrix.

Physical and Chemical Properties

Particle-Size Distribution

In the matrix, total clay content was highest in the BCrt horizon and decreased with increasing depth to a minimum in the Cr3 horizon (Table 2). In all four weathered bedrock horizons, total clay content in the fracture rind was higher than that in the matrix. In the BCrt and Cr1 horizons, total clay contents were highest in the fracture rind, whereas in the Cr2 and Cr3 horizons, they were highest in the fracture coating. Fine clay contents showed vertical and lateral trends identical to those for total clay (Table 2).

Gravel/(sand + silt) ratios were calculated to assess the relative proportions of sand- and silt-sized particles generated by physical weathering, or fragmentation, of gravel-sized primary mineral grains in the parent bedrock. In the matrix, gravel/(sand + silt) ratios increased with increasing depth (Table 3). In all four weathered bedrock horizons, gravel/(sand + silt) ratios were lower in the rind than in the matrix. With the exception of the Cr3 horizon, gravel/(sand + silt) ratios in the fracture-fill materials were generally lower than those in matrix but were comparable to those in the rind.

Pedogenic Iron (Fe_d)

In the matrix, pedogenic Fe content was highest in the BCrt horizon and decreased with increasing depth to a minimum in the Cr3 horizon (Fig. 4). Lateral trends for pedogenic Fe content closely paralleled those for total and fine clay content, increasing from a minimum in the matrix to a maximum in the fracture rind or coating. The highest pedogenic Fe contents in the soil-weathered bedrock profile occurred in the matrix (7.7 g kg⁻¹) and rind (9.4 g kg⁻¹) of the BCrt horizon.

Bulk Density, Total Porosity, and COLE

The bulk density of the weathered bedrock matrix was lowest in the BCrt and Cr1 horizons (1.83–1.86 g cm⁻³), and increased to 2.02 g cm⁻³ in the Cr2 horizon and 2.10 g cm⁻³ in the Cr3 horizon (Table 4). In the BCrt and Cr1 horizons, the bulk density of the rind was greater than that of the matrix, whereas in the Cr2 and

Table 2. Particle-size distribution of the soil and weathered bedrock at the study site.

Horizon	Morphologic zone														
	Matrix†					Fracture rind					Fracture coating				
	Gravel	Sand	Silt	Clay‡	Fine clay	Gravel	Sand	Silt	Clay	Fine clay	Gravel	Sand	Silt	Clay	Fine clay
A	29	56	29	15	7	—§	—	—	—	—	—	—	—	—	—
AB	35	58	26	16	7	—	—	—	—	—	—	—	—	—	—
BCrt	26	51	23	^a 26 _a	12	20	46	24	^a 32 _a	15	—	—	—	—	—
Cr1	32	59	25	^a 16 _b	7	19	56	23	^a 21 _b	10	—	—	—	—	—
Cr2	39	64	25	^a 11 _b	5	21	61	26	^a 13 _c	6	—	—	—	—	—
Cr3	55	68	24	^a 8 _b	3	30	62	21	^b 17 _{bc}	8	—	—	—	—	—

† Gravel content expressed as percentage of whole soil or weathered bedrock; sand, silt, clay, and fine clay content expressed as percentage of the fine earth (<2 mm) fraction. ANOVA was conducted for clay content only.

‡ Column values with different superscripts differ significantly ($P \leq 0.05$). Row values with different superscripts differ significantly ($P \leq 0.05$). Particle-size analyses were not replicated for the A and AB horizons (i.e., $n = 1$). Consequently, clay content values for these two horizons were not included in the ANOVA.

§ — = not applicable.

Table 3. Gravel/(sand + silt) ratios for the soil and underlying weathered bedrock at the study site. Ratios were calculated using gravel, sand, and silt contents expressed on a whole soil or whole weathered bedrock basis.

Horizon	Depth	Morphologic zone			
		Matrix	Fracture rind	Fracture coating	Fill
	cm				
A	0-2	0.5	—†	—	—
AB	2-7	0.6	—	—	—
BCrt	7-17	0.5	0.4	—	0.4
Cr1	17-45	0.6	0.3	—	0.4
Cr2	45-88	0.7	0.3	—	0.4
Cr3	88-125+	1.1	0.6	—	1.7

† — = not applicable.

Cr3 horizons the reverse was true. In the weathered bedrock matrix, COLE values were highest in the BCrt horizon (0.029), and decreased with increasing depth to a minimum of 0.017 in the Cr3 horizon (Table 4).

Organic C

The OC contents were $<10 \text{ g kg}^{-1}$ in all soil and weathered bedrock horizons, but exhibited distinct trends with depth and by morphologic zone that closely paralleled root distributions (Fig. 5 and Table 1). The OC decreased with increasing depth in the matrix and fracture rind, but remained high and relatively constant in the fracture-coating material.

DISCUSSION

Matrix, Matrix Rind, and Fracture Rind

Chemical Weathering

The increase in feldspar pitting and biotite expansion upward in the bedrock profile and laterally towards joint fractures suggests that the higher clay and Fe_d contents near joint fractures and the soil-weathered bedrock interface (Table 2 and Fig. 4) are in part the result of greater chemical weathering. In weathered granitic bedrock, water flows downward from the overlying soil into the matrix and, under saturated conditions, preferentially into joint fractures (Johnson-Maynard et al., 1994; Graham et al., 1997; Frazier, 1997). Plant roots are most abundant near the soil-weathered bedrock interface and in the fracture-rind and fracture-void spaces, and can promote weathering via their decomposition products, exudates, and microbial associates (Spyridakis et al., 1967; Schnitzer and Kodama, 1976; Tan, 1978; Berthelin, 1983; Leyval and Berthelin, 1991). Thus, the most chemically weathered zones in the bedrock are those that are most directly and frequently exposed to percolating water and penetrating roots, both vertically downward from the overlying soil and laterally as water and roots move inward from the joint fractures.

Clay Illuviation and Macroporosity

The presence of illuvial void coatings and infillings throughout the matrix indicates that clay translocation is not limited to fractures. Water flow through the matrix also redistributes clay in suspension. Depth functions

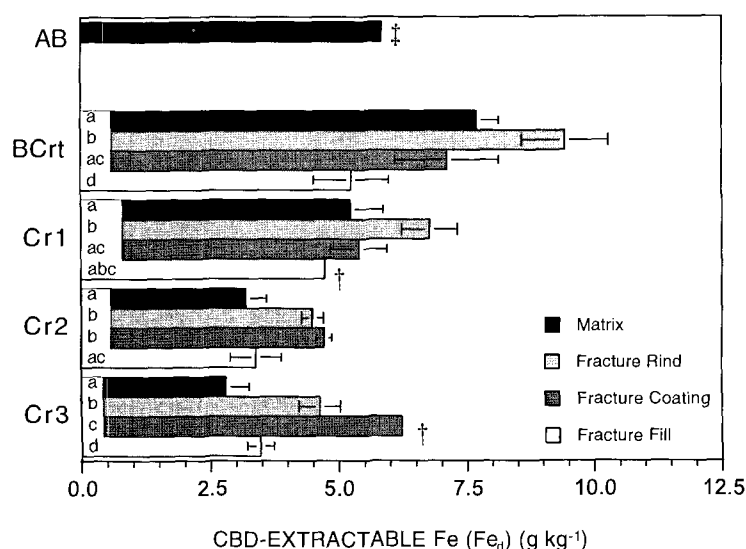


Fig. 4. Citrate-bicarbonate-dithionite extractable Fe (Fe_d) content of the soil and weathered bedrock at the study site. Value bars for a given horizon that are inscribed with different letters differ significantly ($P \leq 0.05$). Error bars represent one standard deviation about the mean. † Error bars are not visible on scale shown. ‡ $n = 1$, and therefore standard deviation could not be calculated.

of total and fine clay (Table 2) and void ferriargillans (Fig. 3d) indicate that the BCrt is the main illuvial horizon. Comparatively lower total and fine clay contents of the A and AB horizons show them to be eluvial and a source of clay for the BCrt horizon. The predominance of rock-controlled structure in the matrix suggests that the BCrt horizon attains its argillic character before it undergoes significant structural development. Thus, illuvial processes involved in the formation of the BCrt are facilitated largely by pedogenic porosity generated through weathering and root pedoturbation, and not by structural, interpedal porosity, as is the case in many soils.

In the matrix, quantities of void infillings and planar macrovoids are inversely related as a function of depth (Fig. 3a and 3d). These data suggest that planar macrovoids in the rock fabric become progressively clogged with illuvial clay over time, rendering them less effective for water flow, a phenomenon also observed in saprolites of the southeastern USA (Vepraskas et al., 1991; Williams et al., 1994; Vepraskas et al., 1996). Consequently, root channels become increasingly important as conduits for clay translocation in the upper regions of the profile as the BCrt horizon develops. Root channels, which compose >90% of the macrovoid volume in the

BCrt matrix (Fig. 3a), also accumulate illuvial clay, but root growth creates new channels or reopens old ones (Fig. 2b), so they facilitate clay translocation when other types of voids have been clogged.

The color and elemental composition of void argillans have been used to determine the origins of illuvial clay in granitic bedrock (Nettleton et al., 1968; Chartres and Walker, 1988; Graham and Franco-Vizcano, 1992). Chartres and Walker (1988) found that the elemental composition of illuvial clay contained in deep weathered bedrock horizons differed from that of the soil clay and illuvial clay in shallower bedrock horizons, suggesting that the deeper clay deposits had been translocated from sources other than the soil. Similar conclusions were made in this study based on color rather than elemental composition. Ferriargillans in the BCrt matrix have redder hues than ferriargillans in the matrices of underlying Cr horizons, suggesting that clay translocation occurred from different sources in the regolith. Most of the illuvial clay in the BCrt matrix is probably derived from the overlying soil, while illuvial clay in the underlying Cr horizons appears to originate largely from localized translocation of clay generated from in situ weathering of biotite, hornblende, and plagioclase feldspars. These minerals are likely sources of illuvial clay during early

Table 4. Particle density, bulk density, total porosity, and COLE† values of the soil and underlying weathered bedrock at study site.

Horizon	Depth cm	Matrix						Rind			
		Particle density g cm ⁻³	Bulk density		Porosity %	COLE		Particle density g cm ⁻³	Bulk density		Porosity %
			<i>n</i>	mean§ g cm ⁻³		<i>n</i>	mean		<i>n</i>	mean§ g cm ⁻³	
A	0–2	2.68	13	1.55	42.3	nd‡	nd	–‡	–	–	–
BCrt	7–17	2.67	10	^a 1.83 _a	31.2	3	0.029	2.65	18	^a 1.87 _a	29.5
Cr1	17–45	2.68	6	^a 1.86 _a	30.7	1	0.023	2.66	19	^b 1.92 _a	27.8
Cr2	45–88	2.70	9	^a 2.02 _b	25.1	2	0.019	2.67	19	^b 1.83 _b	31.6
Cr3	88–125+	2.71	5	^a 2.10 _b	22.5	2	0.017	2.68	15	^b 1.89 _b	30.0

† COLE, coefficient of linear extensibility.

‡ nd = not determined; – = not applicable.

§ Column values with different subscripts differ significantly ($P \leq 0.05$). Row values with different superscripts differ significantly ($P \leq 0.05$).

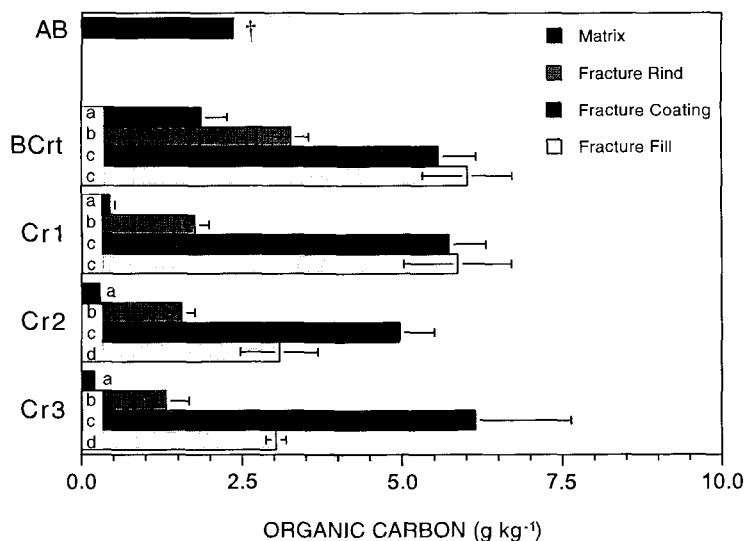


Fig. 5. The OC content of the soil and weathered bedrock at the study site. Value bars for a given horizon that are inscribed with different letters differ significantly ($P \leq 0.05$). Error bars represent one standard deviation about the mean. † Error bars are not visible on scale shown.

stages of pedogenesis, as they are relatively easily weathered and abundant in granitic rock. The colors indicate a lower Fe oxide content and reflect the lesser degree of weathering in the Cr horizons.

In all four weathered bedrock horizons, clay contents are higher and the ratio of fine clay/total clay is higher in the fracture rind than in the adjacent matrix (Table 2), suggesting that the rind is relatively enriched with illuvial clay. As with weathering, illuvial enrichment of the rind appears to be facilitated by the movement of water down joint fractures. The most probable source of illuvial clay in the rind is the overlying soil, which generally has lower clay contents and lower fine/total clay ratios than the rind (Table 2).

Mineral Grain Fragmentation and Localized Fabric Disruption

Micromorphologic observations and gravel/(sand + silt) ratios (Table 3) indicate that the abundance of fragmented mineral grains and the degree of mineral grain fragmentation increase upward in the profile. The increase in grain fragmentation coincides with an increase in the abundance and thickness of void ferriargillans (Fig. 3d). Mineral grains separated by planar voids that are infilled with illuvial clay often exhibit clustered-related distribution patterns with boundaries that accommodate adjacent grains in the clusters (Fig. 2a). These features are usually associated with areas of locally disrupted rock-controlled fabric, and become more abundant upward in the profile. Thus, we speculate that shrink-swell activity of illuviated clay within planar voids and microcracks plays a major role in grain fragmentation and localized disruption of rock-controlled fabric. Rock-controlled fabric predominates throughout most of the matrix, however, indicating that shrink-swell activity generally has not been sufficient to induce the development of soil structure, except in parts of the BCrt matrix where weak secondary blocky structure exists. Similar observations and conclusions as to the

role of illuvial clay in physical weathering of granitic bedrock have been made in southern California (Nettleton et al., 1968; Graham et al., 1994) and in southeastern Australia (Dixon and Young, 1981; Chartres and Walker, 1988).

Illuvial clay content and grain fragmentation trends similar to those that occur with depth also occur laterally from the rind to the matrix. In the Cr2 and Cr3 horizons, void ferriargillans (Fig. 3d and 3e) and fragmented mineral grains are more abundant, and the degree of grain fragmentation and localized fabric disruption are more advanced, in the matrix-rind zone than in the matrix. We interpret that the higher illuvial clay contents are responsible for the greater degree of grain fragmentation and fabric disruption in the matrix-rind zone. In the BCrt and Cr1 horizons, fragmented mineral grains and the degree of mineral grain fragmentation follow the same trends as described above, but illuvial clay contents in the matrix rind and matrix of these horizons are similar (Fig. 3d and 3e). These data and observations suggest that roots play a greater role in fabric disruption in these horizons than in the underlying Cr2 and Cr3 horizons where roots are generally restricted to the fracture rind and the fracture void space (Table 1).

Fabric Reorganization

Except for the Cr3 horizon, the rock-controlled fabric that predominates in the weathered bedrock matrix and matrix-rind zones does not occur within the fracture rind. The common and distinguishing features of the rind fabric in the Cr2, Cr1, and BCrt horizons, such as the parallel-referred orientation pattern of platy and elongate mineral grains (Fig. 2c and 2e), contorted mica pseudomorphs (Fig. 2c and 2d), and the granostriation of clay and mica pseudomorphs around quartz and feldspar grains, suggest that the fracture rind in these horizons has been laterally compressed. We hypothesize that swelling of pedogenic clay in the weathered bedrock causes joint fractures to close upon wetting, and that

the resulting compressive and shear stresses are responsible for forming the fabric exhibited in the rind in the BCrt, Cr1, and Cr2 horizons. Although the mineral structures of vermiculite and kaolinite, the two main clay minerals present in the weathered bedrock (Frazier, 1997), expand only very slightly upon wetting, shrink-swell is largely the result of the loss and gain of water in the microporous network formed by linked phyllosilicate crystallites, rather than interlayer expansion (Wilding and Tessier, 1988). Measured COLE values indicate that there is sufficient clay present in the weathered bedrock matrix to cause as much as a 3% increase in volume because of swelling (Table 4). Additionally, the aperture of joint fractures in a nearby (≈ 0.5 m) weathered granitic bedrock decreased considerably, to the point of total closure at some depths, when wetted with an aqueous tracer suspension (Frazier, 1997). Apparently, the rock-controlled fabric in the matrix contains enough clay to expand upon wetting, but is sufficiently rigid to compress the rind material rather than deform when fracture sidewalls converge.

Bulk density trends provide further evidence that lateral compression of the rind has taken place (Table 4). Studies have shown that the bulk density of granitic rock decreases as it weathers (Nettleton et al., 1970; Clayton et al., 1979; Chartres and Walker, 1988; Graham et al., 1997). We found this to be true for the bedrock matrix, in which bulk density decreases upward from the Cr3 to the BCrt horizon. Because weathering also increases towards joint fractures, a similar decrease in bulk density would be expected from the matrix to the fracture rind. While this is the case in the Cr2 and Cr3 horizons, the bulk density of the fracture rind in the BCrt and Cr1 horizons is slightly greater than that in the adjacent matrix. Thus, we conclude that shrink-swell processes have compacted the rind materials in these most weathered, clay-enriched horizons.

Compaction may explain the relative paucity of void ferriargillans in the Cr1 and Cr2 rinds (Fig. 3f), despite the fact that they have higher clay contents and higher fine/total clay ratios than the adjacent matrix (Table 2). Apparently void ferriargillans are being disrupted faster than they are created in these horizons. The opposite appears to be true in the BCrt rind, where void ferriargillans are more abundant than in the adjacent matrix (Fig. 3d and 3f). Apparently, the rate of ferriargillan formation in this horizon exceeds the rate of disruption.

Lateral compression may also explain why roots rarely extend from joint fractures laterally into the rind of the BCrt, Cr1, and Cr2 horizons. Root penetration could be limited by the dense, compacted nature and vertically trending fabric of the rind in these horizons. Based on the abundance of vertically trending root channels in the rind (Fig. 2e), however, it appears that root growth downward through the rind is not restricted.

In the Cr3 horizon, the rind fabric does not differ significantly from that in the adjacent matrix-rind zone. Additionally, macroporosity, total porosity, and the volume of void ferriargillans are greater in the rind than in the adjacent matrix-rind zone (Fig. 3 and Table 4). These data and observations suggest that the Cr3 rind

has not undergone lateral compression and compaction, and that the matrix-rind zone develops along the margins of joint fractures prior to fabric reorganization.

In weathered bedrock horizons where laterally compressed rind fabrics exist (i.e., BCrt, Cr1, and Cr2 horizons), there appears to be very little mixing of the material in the rind and matrix rind, as the boundary between the two zones is usually sharp and well defined (at $20\times$ magnification; Bullock et al., 1985). Therefore, we deduced that thickening of the rind fabric occurs via two mechanisms. First, more resistant grains at the matrix rind-rind boundary (Fig. 2c) become dislodged and incorporated into the rind fabric by shrink-swell generated stresses when fracture sidewalls converge. Localized shrink-swell activity of illuvial clay within intra- and intergranular planar voids (Fig. 2c) may loosen grains from the rock-controlled matrix rind fabric, allowing them to be more easily dislodged and incorporated into the rind fabric when fracture sidewalls converge. Second, grains weakened by chemical weathering at the matrix rind/rind boundary may be broken and incorporated into the rind by similar stresses. Fragmentation and subsequent incorporation of grains into the rind via these mechanisms is probably responsible for the lower gravel/(sand + silt) ratios in the rind relative to the matrix (Table 3).

Structural Development

In the BCrt horizon, some fracture-rind samples have crack or fissure microstructure, usually together with incipient subangular blocky microstructure (Fig. 2f). The microstructural units are bounded by what appear to be desiccation cracks, attesting to the important role of shrink-swell processes in the development of subsoil microstructure. We speculate that formation of the laterally compressed, vertically trending fabric that occurs in most samples of the BCrt rind (Fig. 2e) precedes the development of crack, fissure, and subangular blocky microstructures on the basis that the rock-controlled fabric that initially exists along joint fracture margins would need to be disrupted before substantial microstructural development could begin. The matrix rind-rind boundary is less pronounced (at $20\times$ magnification) where the BCrt rind has crack and subangular blocky microstructure, suggesting that fabric reorganization and microstructural development proceeds inward from the rind, following the weathering front and water infiltration pathways. Weak secondary soil structure develops concurrently at the soil-weathered bedrock interface (Table 1) and proceeds downward following the vertical weathering front and water infiltration pathways.

Fracture Coating

The ratio of fine clay/total clay is higher in the fracture coating than in overlying soil horizons (Table 2), suggesting that the fracture coatings are illuvial. In the BCrt and Cr1 horizons, both fine and total clay contents are lower in the fracture coating than in the rind (Table 2). The opposite relationship exists in the Cr2 and Cr3

horizons. Additionally, the fine/total clay ratio of the fracture coating in the BCrt horizon is lower than that for fracture coatings in deeper horizons. These data suggest that clay has been eluviated from the rind surface in the BCrt and Cr1 horizons and deposited as a coating on rind surfaces in the underlying Cr2 and Cr3 horizons. The illuvial nature of the rind surface in the lower profile is especially apparent in the Cr3 horizon where the clay content in the fracture coating is more than double that in the matrix (Table 2).

The smoothed fracture coatings in the upper part of the profile appear to result from the same compressive forces that reorganized the fabric in the rind. Thoma et al. (1992) found that mineralized coatings on the surfaces of fractures in weathered tuff had a significant effect on the migration of water between fractures and the surrounding matrix. In this study, the presence of smoothed coatings, or pressure faces, in the upper part of the profile, as well as the vertically trending fabric and low lateral pore continuity of the rind zones, may inhibit the flux of water into the rind and surrounding matrix in an analogous manner. As a result, water may move relatively rapidly and frequently across the BCrt rind surfaces, eluviating clay in the process.

Fracture Fill

The data collected during this study do not conclusively indicate the origin of the fracture-fill materials. Fill materials in the BCrt and Cr1 horizons have particle-size distributions similar to those for overlying soils (Table 2), suggesting that they may have sloughed into joint fractures from above. Conversely, fill materials in the Cr2 and Cr3 horizon have particle-size distributions that more closely resemble those in the adjacent matrix, suggesting that they may be derived from the sloughing of fracture sidewalls. The vertical platy structure of some fill materials is consistent with the lateral compression hypothesis invoked to explain fabric reorganization in the rind.

SUMMARY

Clay illuviation and shrink-swell processes play critical roles in the pedogenic transformation of this granitic bedrock. Water infiltrates the matrix vertically at the soil-weathered bedrock interface and laterally from bedrock joint fractures, resulting in greater weathering and illuvial clay deposition upward in the bedrock profile and adjacent to joint fractures. Swelling of illuvial clay deposited within pedogenic voids causes grain fragmentation and localized disruption of rock-controlled fabric. Swelling forces manifested on a larger scale cause fracture sidewalls to converge, resulting in compaction and reorganization of rock-controlled fabric along fracture margins. Both localized and large scale shrink-swell processes and their morphologic effects are most substantial upward in the profile and adjacent to joint fractures where weathering and clay content are greatest. Microstructural development is facilitated by the reorganization of rock-controlled fabric along fracture margins, and moves inward following the weathering front

and water infiltration pathways. Weak subsoil structure develops concurrently at the soil-weathered bedrock interface and proceeds downward following the vertical weathering front and water infiltration pathways.

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ERRATUM

Lateral Podzolization in a Granite Landscape

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Soil Sci. Soc. Am. J. 64:1434–1442 (July–August 2000).

On p. 1438, the second footnote in Table 2 should indicate that acid digestion was only for G-2; i.e., ‡Only for data of G-2 (Keilen, 1978): acid digestion with HF/HClO₄ at 210°C (Jackson, 1958).

On p. 1438–1439, Tables 2 and 3 should not include the pound signs before the values in the tables. In Table 2, there should be a pound sign on the first occurrence of bdl and the corresponding footnote; i.e., #bdl = below detection limit of x-ray diffraction.

On p. 1438, the caption for Fig. 2 should identify profiles G-1 and G-4, rather than G = 1 and G = 4.